

Afik Aharon

Software Engineer · Backend & AI · 3rd Year CS Student

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Backend developer and CS student with hands-on experience building production-grade systems, Generative AI pipelines, and full-stack applications. Proven ability to deliver in high-stakes enterprise environments, debug complex architectures, and learn new technologies rapidly. Eager to contribute to innovative teams and grow.

EDUCATION

B.Sc. Computer Science | [HIT - Holon Institute of Technology](#) | Jan 24 - Expected 26

- Coursework: Data Structures & Algorithms, Data Science, Cloud Infrastructure (AWS), Operating Systems.
- Developing expertise in scalable backend design and high-performance AI architectures.

WORK EXPERIENCE

System Consultant & Backend Developer | [Ness Technologies \(student position\)](#) | Nov 25 - Present

- Backend development and system consulting for mission-critical, large-scale IDF Mainframe enterprise systems.
- Maintain and optimize high-scale architecture ensuring continuous production stability in secure environments.
- Investigate and resolve complex data-flow and architectural issues, requiring deep debugging across distributed systems.

Mainframe Backend Developer | [IDF - Shahar Tech Unit](#) | 2020 - 2023

- Developed and maintained critical backend systems on IBM Mainframe, translating complex business requirements into robust production code.
- Led end-to-end debugging of complex production failures, ensuring zero-downtime stability for core military systems.
- Mentored new developers on system architecture and collaborated cross-functionally on large-scale deliverables - demonstrating ownership and leadership early in career.

PROJECTS

Generative AI for Architectural Change Detection | [Python](#) · [PyTorch](#) · [Hugging Face](#) | Oct 25 - Feb 26

- Built an end-to-end AI pipeline to detect structural changes in architectural floor plans using Computer Vision and GenAI.
- Integrated DETR (object detection), SAM (segmentation), and Stable Diffusion (inpainting) into a single synthetic data generation system - demonstrating complex system design skills.
- Applied precision-recall optimization: identified a 0.30 decision threshold that boosted recall from ~50% to 96.2%, achieving F1 of 0.81.

Intelligent Robotic Arm for Object Sorting | [C++](#) · [Arduino](#) · [React](#) · [Computer Vision](#) | Jun 25 - Oct 25

- Engineered an autonomous robotic arm with Arduino/ESP32 and a Vision AI module for real-time color-based sorting.
- Built a React-based web UI for live monitoring, manual servo control, and calibration - showcasing full-stack capability alongside embedded systems.
- Implemented automatic and manual operation modes, combining hardware control with computer vision algorithms.

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, Java, C/C++, .
Frontend	React.js, HTML/CSS.
Backend & Cloud	Node.js, AWS (pursuing certification), Linux, Git, CI/CD basics, REST APIs.
AI / ML	Generative AI pipelines, LLMs, Agentic workflows (LangChain/LangGraph), RAG, Prompt Engineering, PyTorch.
Languages	Hebrew (Native) · English (Professional)